

Optimal design of hydrogen-based storage with a hybrid renewable energy system considering economic and environmental uncertainties

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ABSTRACT

Hydrogen and electricity derived from renewable sources present feasible alternative energy options for the decarbonisation of the transportation and power sectors. This study presents the utilisation of hydrogen generated from solar and wind energy resources as a clean fuel for mobility and backup storage for stationary applications under economic and environmental uncertainties. This is achieved by developing a detailed techno-economic model of an integrated system consisting of a hydrogen refuelling station and an electric power generation system using Mixed Integer Quadratic Constrained Programming (MIQCP), which is further relaxed to Mixed Integer Linear Programming (MILP). The model is implemented in the Advanced Interactive Multidimensional Modelling Software (AIMMS) and considering the inherent uncertainties in the wind resource, solar resource, costs and discount rate, the total cost of the three configurations (Hybrid PV-Wind, Standalone PV, and Standalone wind energy system) was minimised using robust optimisation technique, and the corresponding optimal sizes of the components, levelised cost of energy (LCOE), excess energy, greenhouse emission avoided, and carbon tax were evaluated. The levelised cost of the deterministic optimisation solution for all the configuration ranges between 0.0702 \$/kWh to 0.0786 \$/kWh, while the levelised cost of the robust optimisation solution ranges between 0.07188 \$/kWh to 0.1125 \$/kWh. The proposed integration has the advantages of affordable hydrogen and electricity prices, minimisation of carbon emissions and grid export of excess energy.

1. Introduction

While the Industrial Revolution improved the quality of life due to new opportunities and economic growth, the accompanying anthropogenic activities have brought increased greenhouse emissions to the atmosphere, causing severe climatic challenges [1,2]. The rise in the Earth's average surface temperature can be attributed to greenhouse gases, which contribute to the depletion of the planet's ozone layer. This layer encompassing the whole Earth plays a crucial role in safeguarding life by absorbing ultraviolet-B (UV-B) radiation emitted by the sun. The elevated surface temperature engenders several adverse effects on human health, including but not limited to global warming, rising ocean levels and acidification, climate change, droughts, floods, and other related consequences [3]. The present energy economy relies on fossil fuels, the availability of which is experiencing a rapid decline. Moreso, the concentration of atmospheric CO₂ has surpassed 400 ppm in recent times, having previously remained below 300 ppm before the year 1950 [4]. The transport and electricity sectors are the most significant contributors of greenhouse gases to the atmosphere, as more than 94 % of

the fuel used for transportation is fossil-based, while about 79 % of electricity generation comes from coal and natural gas [5]. Coal, a fossil fuel with a very high carbon footprint, dominates the Republic of South Africa's electricity generation mix, providing 80 % of the total electricity generated in the country [6]. Due to these reasons, the two sectors have been the focus of heavy decarbonisation through the use of cleaner energy sources [2,7,8].

Renewable energy resources (RES) are increasingly deployed in powering electric grids and the production of hydrogen fuel for refuelling stations. These energy infrastructures have the advantages of modularity and relatively affordable installation/maintenance costs [2]. RES, especially solar and wind energy, are primarily abundant in nature, naturally replenishable, friendly to the environment and a suitable alternative to the fossil-dominated economy [9]. Hydrogen generated from renewable energy is a clean fuel and a great prospect for reducing fossil-based fuels in the transport industry. However, a notable distinction between RES and traditional energy sources is in the inherent limitation of dispatchability. Also, unlike conventional sources, RES cannot be accurately predicted due to their output being characterised

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by randomness, making it challenging to forecast precisely.

The complementary operation of solar PV and wind turbine have demonstrated their competence to solve the drawbacks of a renewable energy system in terms of performance, reliability and cost [10–12]. To further improve the performance of the hybrid system, energy storage is incorporated to balance the intermittent and stochastic nature of the power supply. Among different storage methods, battery and hydrogen storage stand out due to their flexibility in operation and they can be deployed to any location. Battery storage has lower cost, high power density (quickness to respond to power fluctuations) and high round-trip efficiency. However, it cannot store energy for a long time because it has self-discharge rate, relatively low energy density and bulkiness. Hydrogen storage offers the merit of compactness, low rate of self-discharge and very high energy density. Continued market penetration of hydrogen-based storage systems will further reduce the unit cost of hydrogen.

Integrating wind-PV and hydrogen technologies significantly facilitates the shift from economies reliant on fossil fuels. In terms of the advantages associated with the utilisation of hybrid systems, they have the potential to enhance the quantity of dispatchable renewable energy generation and improve the dependability of energy access in rural areas [9]. Moreover, using hydrogen-powered cars has the potential to mitigate carbon emissions, leading to a reduction in environmental pollution. This can be achieved by utilising RES for hydrogen production and establishing an extensive network of hydrogen refuelling stations to ensure convenient accessibility [13]. Furthermore, these can be accomplished through favourable governmental policies, financial investments, and optimal techno-economic planning and design. Recently, there has been a significant rise in research efforts focused on exploring the most efficient hybrid system design for power production and hydrogen refuelling. Various methodologies have been employed to identify the optimal configuration of these systems with and without uncertainties. To reduce the carbon footprint of the diesel bus fleet, Coppitters et al. [14] proposed an integrated energy system where hydrogen fuel produced from hybrid wind and solar system is added with diesel fuel to power a fleet of buses and Robust Design Optimisation (RDO) which considers the uncertainties in the economic, technical and environmental parameters is used to optimise the configuration of photovoltaic and wind turbine powered hydrogen refuelling system and fleet of diesel bus. Levelised driving cost(LCOD) and intensity of carbon (C.I), both subject to uncertainty, are minimised. 54 % of the fleet of diesel-fuelled buses would be replaced by hydrogen-fuelled buses, which would result in a reduction in LCOD standard deviation(SD) of 36 %, a reduction in C.I mean value of 46 %, and a reduction in C.I standard deviation(SD) of 51 %, but only a slight increase in LCOD mean of 11 %. These results clearly show that the system stands to perform better economically and environmentally. In another study, Coppitters et al. [15] proposed a grid-connected solar PV with battery and hydrogen storage, which was optimised using multiple scenarios of robust optimisation and then compared with stochastic optimisation which depends on the cumulative density functions. The solar PV configuration has the lowest LCOE (269 €/MWh) and lowest self-sufficiency (less or equal 30 %) ratio, which is without the risk of power cuts and blackouts due to its reliance on the grid. Hybridising solar PV and hydrogen/battery storage provides a more competitive solution over full-grid reliance, reducing the risk of blackouts and power cuts. Furthermore, [16] conducted a surrogate-assisted robust design optimisation to obtain the optimal levelised cost of hydrogen of a solar PV – electrolyser system in some commercial cities around the globe under several uncertain parameters. It was deduced that the location with the highest annual solar irradiance has the lowest levelised cost of hydrogen (LCOH) of 6.6 €/kg and a standard deviation (SD) of 0.72 €/kg. It was also reported that the uncertainty in the discount rate caused the highest variation in the unit cost of hydrogen. C.A. Wankouo Ngouleu et al. [9] provided a comparative assessment of four different *meta-heuristic* optimisation methods for the optimal sizing of hybrid solar PV and wind turbine with

battery and hydrogen storage for three typical loads in a location in Cameroun. Six different configurations are sized while minimising the net present cost and increasing the reliability using imperialist (I), cuckoo-search (C.S), artificial-bee-colony (A.B.C) and big-bang-big crunch (B.B.B.C) algorithms. It was revealed that the cuckoo-search algorithm performs better than the other three algorithms, and off-grid solar PV/wind/battery outplays others in terms of cost and reliability. Nevertheless, the hydrogen-based storage is more reliable and environmentally-friendly. It was further stated that with the reduction of cost and improvement of the efficiency of the fuel cell and electrolyser, hydrogen storage has better prospects.

M.M. Samy et al. [17] presented an optimal design of a hybrid energy system consisting of solar PV, wind-turbine, and fuel-cell connected to an unreliable grid in a tourist location in Egypt. The model of the sub-systems coupled with the power management system were developed, and Hybrid Firefly and Harmony Search Optimisation (HSO) techniques were used to obtain the optimal size of each component for minimal net present cost. Particle Swarm Optimisation technique was used to validate the results. The results reflect that the presented system is more viable than the grid. The LCOE is 0.0628 \$/kWh, while the tariff cost in Egypt is 0.1 \$/kWh. C. Ghenai et al. [8] presented a modelling, simulation and optimisation of solar PV and fuel-cell to satisfy the electrical load of 150 households in the Sharjah Emirate, UAE. Accumulation of dust and atmospheric temperature, impacting the solar PV's power-output, are considered in this study. The developed system should not just supply the residential community's daily and annual energy needs but also boost the amount of clean energy in the power mix, lessen the quantity of dump, have a low LCOE, and have low environmental impacts. The distributed hybrid energy system(DHES) generated 52.00 % of the power from solar PV, while 48.00 % came from fuel cells. 80.70 % of the total electricity produced by the photovoltaic-hydrogen-fuel-cell hybrid system is used to satisfy all the residential community's AC demands, leaving only a small amount of AC primary load unmet (0.08 %), 14.08 % is used as the input DC power for the electrolyser utilised to produce hydrogen, 3.30 % is lost in the DC-AC inverter, and 1.84 % is dumped. With a LCOE of 0.145 \$/kWh, the suggested off-grid hybrid-renewable power system is economically feasible, environmentally sustainable, and has a 40.2 % renewable contribution with no carbon dioxide emissions. A. Abdollahipour and H. Sayyaadi [18] proposed a study to investigate the performance of Proton Membrane Fuel Cell and Proton Membrane Electrolyser when they are coupled with solar PV and critical factors such as efficiency, the power-output and LCOE are evaluated. The electrolyser and fuel cell governing equations were solved using COMSOL MULTIPHYSICS. Additionally, a multi-objective optimisation was carried out with the dual goals of minimising the levelised cost of the system while simultaneously maximising its output power and efficiency. The power generation system's efficiency and output power were 0.323 and 1.802 kW/m², respectively, while the levelised cost of energy of the system's lowest value was 0.498 \$/kWh. Assuming that all of the hydrogen fuel produced by the electrolyser is used to power a fuel cell, this analysis does not consider hydrogen storage. M.M. Nasiri et al. [19] developed an optimisation model for the optimal sizing and investment of a building's (hospital) distributed energy system. A deterministic and robust model to maximise the revenue/energy savings and minimise the costs/carbon emission cost were solved using CPLEX solver in GAMS software. The findings demonstrated that the uncertainties brought on by the degree of conservatism increased the component sizes. The authors concluded that making a trade-off between the mean and standard deviation (SD) of the expenses and the percentage of unfulfilled requests is the key to decision-making under uncertainty conditions for energy systems, notably a hospital. As a result of the study's findings, the deterministic model is recommended. Khiareddine et al. [20] presented a technical and economic optimisation model of an autonomous hybrid renewable energy system consisting of solar PV, wind-turbine, hydrogen and battery-based storage system. The energy system is connected to an induction motor and centrifugal pump

for an agricultural station in a remote location in Tunisia. The research objective is to minimise the system's cost, maximise the system's reliability and provide strategic support for power management of the excess energy. The outcomes emphasise photovoltaic and wind turbine hybridisation's crucial role in lowering system costs. Additionally, it should be emphasised that the hydrogen chain extends battery life by limiting its frequent use. The optimisation sizing algorithm provides every hybrid system configuration that may be used for pumping systems and any autonomous load located anywhere in the world.

A. Maheri et al. [21] introduced a combined configuration-size optimisation formulation which enables the solution of just one optimisation problem to determine the best configuration and the best size for each component. For both on-grid and freestanding systems, single and multiple objective case studies are specified employing wind turbines, solar panels, battery-banks, fuel-cell, electrolyzers, and diesel generators (DGs) as potential components. A non-dominated sorting-genetic-algorithm (NSGA-II) and a genetic-algorithm (G.A) are created to handle the optimisation challenges. The reproduction operators in both algorithms are carefully constructed for robust exploration at both the size and configuration. Eight lone and multi-objective case studies are done for various RES, objectives, and restrictions. To advance the technology of low carbon emission vehicles, W. Hu et al. [22] proposed a combined station for hydrogen refuelling and battery swapping powered by wind turbines and grid. A combined distributionally robust optimisation/stochastic optimisation was proposed to accommodate the different uncertainties in the wind resource, electricity prices and hydrogen demand to minimise the overall operational costs of the station. The results show that, with a high degree of confidence, the proposed technique's total operational cost is 4.40 % less than that of the hybrid stochastic-robust optimisation method, effectively addressing the conservatism of solutions. The waiting time for consumers and the station's operating costs can be decreased with an appropriate battery swap schedule. Zahra Salehi and Imam Gholaminezhad [23] studied the modelling of reversible-solid-oxide cells based on experimental set up which works by using a multi-objective modelling genetic-programming-algorithm (G.A). System voltage is the output parameter, whereas current density, temperature, and fuel mixture are input parameters. The resilience of the resulting models was tested using a Monte-Carlo simulation method for various levels of uncertainty. Statistical indicators like cumulative-distribution-functions (C.D.Fs) and probability-density-functions (P.D.Fs) are used to demonstrate the durability of the ideal G.P-based model's durability. Statistics indices for varying levels of uncertainty from 1 to 10 % are used to demonstrate the robustness of the optimal trade-off model. According to the recommended model, the ideal operating condition has an intermediate temperature of 762.00 °C and a fuel mixture of around 29.00 % H₂, 25.00 % H₂O and 14.00 % CO. A. Kashefi Kaviani et al. [24] presented a study on the optimal design of combined wind-solar PV, electrolyser and fuel-cell to meet power demand reliably. A new variation of the Particle Swarm Optimisation (P.S.O) algorithm minimises system costs over 20 years of its operation, subject to reliability constraints. The likelihood of three primary hybrid system components, a solar array, a wind-turbine generator, and a DC-AC converter failing, is considered. The findings demonstrate how component dependability and cost are directly impacted by outages, i.e., outages necessitate a more extensive generating system to provide the load with an acceptable level of reliability. S. Nistor et al. [25] used a test project (Island-Hydrogen) in the U.K as a trial case to investigate and examine the short- to-medium-term viability of hydrogen fuelling with onsite hydrogen fuel production. To address the current case for hydrogen-fuelled transportation without relying on decreases in future technological costs, the economic performance of the existing hydrogen refuelling technology (both P.E.M and Alkaline-electrolysers) was assessed. The efficiency of a refuelling location in terms of providing for consumers is then assessed based on the principal

energy source (wind power versus grid electricity). The findings indicate that if the return-on-investment term is longer than ten years for P.E.M electrolyzers and five years for Alkaline electrolyzers, hydrogen fuel can compete with petrol on a G.B.P/K.G basis. Additionally, they also demonstrated that capital cost subsidies, reflected in some R&D sponsorship programmes, make both PEM and Alkaline-technologies less expensive than the equivalent price of petrol, indicating that more attention should be placed on commercialising R&D-supported innovations because they offer commercial benefits. The study demonstrated that a station connected to the grid and the wind is preferable since it can service more clients and ensure that there is never a hydrogen shortage.

Numerous studies in the existing literatures reveals that the optimal sizing of solar PV-wind energy systems infrastructures for power generation as well as hydrogen production and refuelling for mobility across diverse locations are frequently conducted independently, with minimal studies exploring their integration. Wind and PV energy systems and other infrastructure are crucial to support the two stations (power generation station and hydrogen refuelling station), and more cost and useable space is incurred if they are to be set up independently. Moreover, alternatives to fossil-based vehicles are now in the early stages of their development. Hence, accurately forecasting the future penetration rates of cars poses a considerable challenge, and investors face a comparatively elevated level of risk when investing in a single station [22]. However, opting for hybridising power and storage systems based on renewables, which has already penetrated the market with green hydrogen refuelling stations, can substantially mitigate risk and reduce the cost of green hydrogen. Moreover, there is a paucity of literature on considering uncertainties in determining the optimal configurations of this energy infrastructure. Most studies conducted in this field have operated under the assumption of fixed parameters. However, this approach can result in designs susceptible to technical, economic, and environmental uncertainties throughout the system's lifespan, resulting in sub-optimal performance.

Based on the motivations presented, this study introduces the optimal design and sizing of an integrated renewable power system and onsite hydrogen refuelling station, considering economic and environmental uncertainties. In order to have a framework that is similar to reality, this study seeks to assess the economic and technical performance of the combined system under several perturbations using robust optimisation. The objective and contribution of the study are first to introduce the feasibility of an integrated renewable power system and the supply chain of onsite hydrogen refuelling hub to meet daily electric and hydrogen demands. Ideally, as separate entities, the renewable power system consists of a photovoltaic, wind turbine, electrolyser, storage tank and fuel cell, while the onsite hydrogen refuelling station consists of photovoltaic, wind turbine, electrolyser, storage tanks, compressor, dispenser, and pre-cooler. The study seeks to model these two frameworks as an integrated system using MIQCP and further relaxed to MILP to evaluate its economic, technical, and environmental benefits.

AIMMS [26–29], an Algebraic Modelling Language tool, is used to model and optimise the system. With the tool, deterministic and robust optimal sizing is carried out for the different configurations of the combined system and their values were compared. The following section of this paper is organised to accomplish these goals: A thorough theoretical model of the system's components, the formulation of the optimisation problem in MIQCP, and a relaxed version of the problem in MILP are all covered in section 2. The case study, the fixed data sets, and the uncertain data sets are highlighted in Section 3. The numerical simulations, findings and discussions are presented in Section 4. Section 5 provides the study's conclusion.

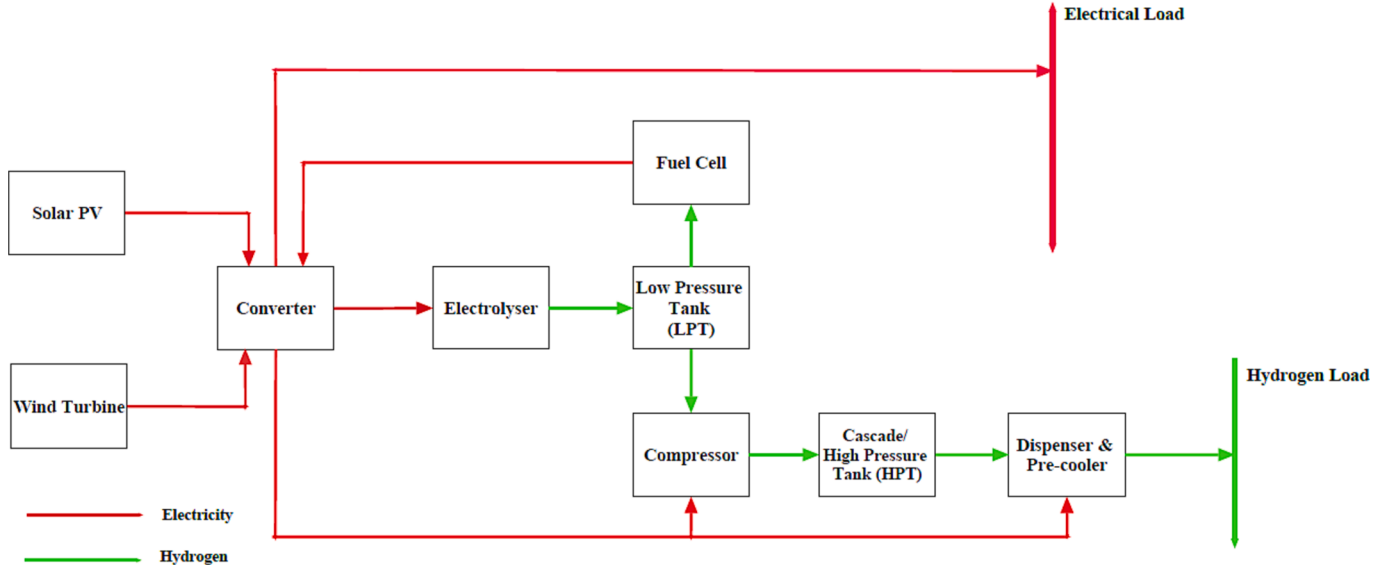


Fig. 1. “Diagram of the Integrated Energy System”.

2. Methodology

2.1. System description

This paper introduces an integrated electricity and hydrogen fuel generation system to satisfy load demand. Fig. 1 illustrates the operational principle of the framework. The electric load demand is met by the power supplied from the power sources (wind turbine and solar PV) and supported by the fuel cell through hydrogen stored in the low-pressure tank (LPT). The hydrogen demand is met through the stored and compressed hydrogen in the cascade. In order to mitigate the risk of overheating in the vehicle tank during refuelling, hydrogen gas is compressed and pre-cooled due to its low density at the standard temperature and pressure. Solar PV and wind turbine convert energy from solar irradiation and wind to serve the load demand. When there is a surplus due to a higher power supply, the excess energy is stored in the low-pressure tank as hydrogen. Hydrogen can be used as long-term energy storage without any significant loss. When there is a shortfall due to higher load demand, the stored hydrogen in the tank is used to meet the hydrogen demand and power the fuel cell to meet the electrical load. The system is configured such that the power released by the fuel cell only supplies the electrical load, compressor, dispenser, and pre-cooler loads when the renewables cannot provide the needed energy. Three different configurations are investigated in this study: when solar PV and wind turbines are acting alone as power sources and when they are working complementarily.

2.2. Solar PV modelling

The solar PV converts the energy from the sun to direct current (DC) electricity. The power generated by the solar PV is given Equ. (1) [4,9,29]. The efficiency (%) and cell temperature of the solar PV is given by Equ. (2) and Equ. (3) respectively.

$$P_{pv}(t) = G(t) * A_{pv} * \eta_{pv}(t) \quad (1)$$

$$\eta_{pv}(t) = \eta_{pv,ref}(1 - \alpha(T_c(t) - T_{ref})) \quad (2)$$

$$T_c(t) = T_{amb}(t) + G(t) \left[\frac{NOCT - 298.15}{800} \right] \quad (3)$$

where $P_{pv}(t)$ denotes the electrical power generated by the solar PV (W), $G(t)$ is the solar irradiance (W/m^2), A_{pv} means the area of the solar panel

(m^2), $\eta_{pv}(t)$ represents the hourly efficiency (%) of the solar panel which varies with the cell temperature, $\eta_{pv,ref}$ denotes the efficiency (%) of the solar panel under test condition, α denotes the temperature coefficient at the maximum power, $(1/k)$, $T_c(t)$ denotes the hourly cell temperature (K), T_{ref} represents the temperature at test condition, K, $T_{amb}(t)$ denotes the temperature of the surrounding (K), NOCT indicates the cell nominal operating temperature (K).

2.3. Wind turbine modelling

The model used to predict the wind turbine's power output, $P_{wt}(t)$ is given below in Equ. (4) [30,31].

$$P_{wt}(t) = \begin{cases} 0 & v_i > v(t) \geq v_o \\ P_r & v_i \leq v(t) < v_o \\ P_r \left[\frac{v(t)^3 - v_i^3}{v_r^3 - v_i^3} \right] & v_i \leq v(t) < v_r \end{cases} \quad (4)$$

where P_r , $v(t)$, v_i , v_r , v_o are the rated power of the wind turbine (kW), the windspeed of the location, turbine cut-in windspeed, turbine-rated wind speed and turbine cut-out wind speed (m/s), respectively.

To estimate the hourly power supply from the wind turbine, the wind speed at the height of the turbine hub is required. The windspeed at the hub must be between the cut-in windspeed and the cut-out wind speed of the turbine for it to generate electricity. The power law is used to extrapolate the speed of the turbine at the turbine hub as given in Equ. (5) [9].

$$V = V_u \left[\frac{h}{h_u} \right]^\beta \quad (5)$$

where V is the wind speed at the hub height (m/s), v_u is the wind speed at the reference height (m/s), h is the hub height (m) while h_u is the reference height (m), β is the roughness factor, which is based on the ground barriers.”

2.4. Electrolyser modelling

Polymer Electrolyte Membrane Electrolyser (PEM-EL) is adopted in this study because it has higher efficiency and operating pressure, ability to produce high purity hydrogen and function under high variability in power supply [32]. The hydrogen exit pressure from the PEM

electrolyser is about 3.5 MPa. Electrolyser efficiency is the ratio of energy between the energy output from the electrolyser in the form of hydrogen and the electrical energy input to the electrolyser [33]. The hourly electrical power consumed by the electrolyser is given in Equ. (6)

$$P_{el}(t) = \frac{\dot{m}_{H_2,d} * HHV_{H_2}}{\eta_{el}} \quad (6)$$

where η_{el} , $\dot{m}_{H_2}$ and HHV_{H_2} are the efficiency of electrolyser (72 %), hourly mass of hydrogen produced (kg/hr) and higher heating value of hydrogen (39.4 kWh/kg) respectively [4].

2.5. Fuel cell modelling

The fuel cell is a device that transforms chemical energy from the fuel into electrical energy. Polymer Electrolyte Membrane fuel cell (PEM-FC) is adopted in this study because it has a promising technology for light duty automobiles and stationary technology applications. It has higher efficiency, DC power generation, minimal emissions, fast response to variable loads and the ability to function under low temperature (60 °C – 100 °C) [24,34]. The hourly electrical power generated by the fuel cell is given by Equ. (7)

$$P_{fc}(t) = \eta_{fc} * \dot{m}_{H_2,d} * HHV_{H_2} \quad (7)$$

where η_{fc} , $\dot{m}_{H_2}$ and HHV_{H_2} are the efficiency of fuel cell (70 %), hourly mass of hydrogen produced (kg/hr) and higher heating value of hydrogen (39.4 kWh/kg) respectively [8].

2.6. Converter modelling

The converter regulates the energy flow between the A.C and D.C bus-bars because it works as an inverter and rectifier. The power rating of the converter is obtained by the system's highest energy demand or supply [30]. The rating of the converter to be used in the system is given in Equ. (8).

$$P_{conv} = \frac{P_{max}}{\eta_{inv}} \quad (8)$$

where P_{conv} , P_{max} and η_{inv} (95 %) are the converter rating, the maximum power, and the efficiency respectively.

2.7. Compressor

According to SAE J2601 which establishes safety limits and performance requirements for gaseous hydrogen cooling and compression, 70 MPa is given to be the maximum pressure in the vehicle tank during refuelling. The electrolyser outlet pressure is about 2 MPa and the compressor compresses it to 90 MPa to accommodate for any pressure loss before and during the refuelling process.

The energy required to compress hydrogen gas adiabatically is given below in Equ. (9) [33].

$$P_{comp}(t) = \frac{T_1 * C_{comp} * ((\frac{P_2}{P_1})^{\frac{\gamma_h-1}{\gamma_h}} - 1) * \dot{m}_{H_2}}{\eta_M * \eta_{comp}} \quad (9)$$

where T_1 , C_{comp} , η_M , η_{comp} and γ_h are temperature of the hydrogen gas at the compressor inlet (293 K), the specific heat of hydrogen at constant pressure (14.304 KJ/kg/k), the efficiency of the compressor (70 %), the efficiency of the motor (90 %) and the specific heats ratio of hydrogen (1.4) respectively.

2.8. Pre-Cooling unit (PCU) & dispenser

Pre-cooling of hydrogen is needed to accommodate for any temperature rise during refuelling process. The energy required to cool hydrogen before refuelling is given in Equ. (10) [22,33]. Based on the

empirical data, an energy of 0.33 kW/kg H_2 is required to pre-cool hydrogen fuel from 15 to –40 °C [25,35].

$$P_{PCU}(t) = 0.33 * \dot{m}_{H_2} \quad (10)$$

The unit is attached to a dispenser, and a single hose dispenser is needed for every 250 kg daily demand [36].

2.9. Low pressure hydrogen tank capacity

To ensure efficient and continuous running of the system, low-pressure tanks are used to store the hydrogen fuel produced. The low-pressure tank (LPT) is located between the electrolyser and compressor to act as buffer when there is a surge in hydrogen demand. The mass balance equations below explain how hydrogen fills and exits the low-pressure tank.

$$m(t) = m(t-1) + X_1(t) * m_{H_2,a}(t) - X_2(t) * m_{H_2,d}(t) \quad (11)$$

$$m_{H_2,a}(t) = \frac{P_{el}(t) * \eta_{el}}{HHV_{H_2}} \quad (12)$$

$$m_{H_2,d}(t) = \frac{P_{fc}(t)}{\eta_{fc} * HHV_{H_2}} \quad (13)$$

$$m_{min} \leq m(t) \leq m_{max} \quad (14)$$

$$X_1(t), X_2(t) \in (0, 1) \quad (15)$$

where $m(t)$, $m(t-1)$, $X_1(t)$, $m_{H_2,a}(t)$, $X_2(t)$, $m_{H_2,d}(t)$, m_{min} and m_{max} are hourly capacity of hydrogen in the tank (kg), capacity of hydrogen in the tank at the previous hour (kg), hourly binary indicator for hydrogen infilling, hourly mass of hydrogen added to the tank through electrolysis (kg), hourly binary indicator for hydrogen exiting the tank, hourly mass of hydrogen removed from the tank through the operation of fuel cell (kg), minimum mass of hydrogen and the maximum/rated capacity for the low pressure tank respectively. It is important to remember that there are minimum and maximum amounts of hydrogen that can be stored. The mass of hydrogen that has been stored cannot possibly be greater than the tank's rated capacity. However, some issues, including a decline in hydrogen pressure, may prevent the extraction of a small portion of the hydrogen (in this case, 5 %) [17,24].

2.10. Cascade/High pressure hydrogen tank capacity

The high-pressure tank (HPT) stores all the compressed hydrogen from the hydrogen compressor. Only the hydrogen demanded at the refuelling point are pre-cooled and dispensed. The size of the cascade depends on the maximum demand which occurs during the peak hour. This is to ensure that there is always a reserve to meet the demand always whenever there is no production of hydrogen due to fault, maintenance or low availability of the renewable energy resources.

$$M_{cascade} = f_s * M_{peak} \quad (16)$$

where M_{cas} , f_s , M_{peak} are the total capacity of the cascade (kg), the scaling factor and maximum demand (kg) respectively.

2.11. Water pump Power/Water tank and cost modeling

The quantity of water needed to produce 1 kg of hydrogen gas is

Table 1
CO₂ equivalence of gasoline and coal [42].

1 kWh	Equivalent Mass of CO ₂ (kg)
Gasoline	0.264
Coal	1.073

Table 2

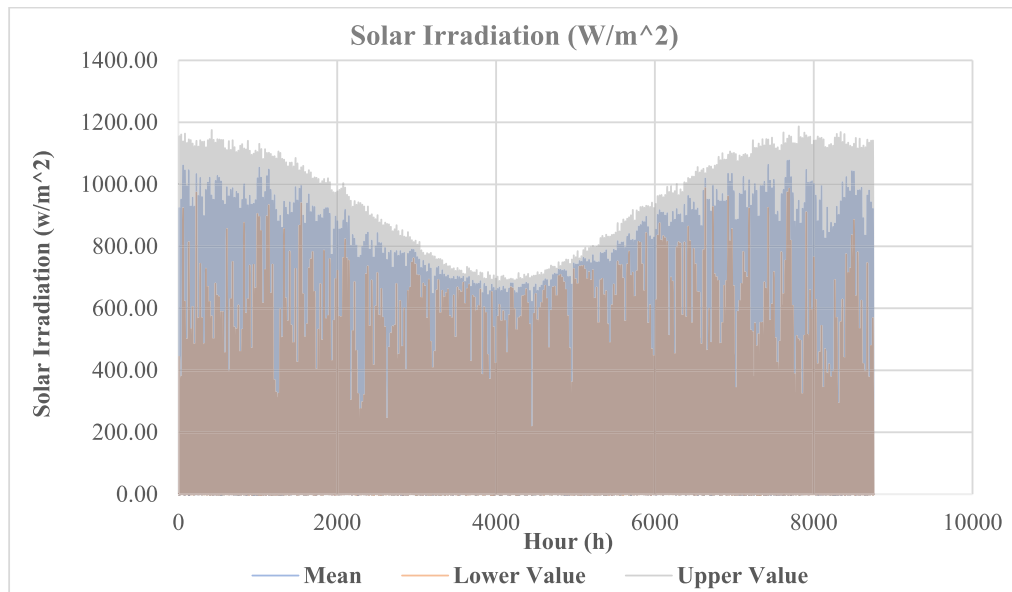
Parameters and their values used in the case study.

S/N	Name of Component	Capital Cost (\$)	Replacement Cost (\$)	O & M Cost (\$/year)	Salvage Cost (\$)	Lifetime (year)	References	CEPCI
1	Wind Turbine	1000/kW	1000/kW	12/kW	0	20	[44,45]	1.37
2	Solar PV	2000/kW	2000/kW	20/kW	0	20	[44]–[46]	1.37
3	Converter	625/kW	625/kW	12.5/kW	417/kW	15	[47,45,38]	1.37
4	Fuel Cell (PEM-FC)	2500/kW	888/kW	0	0	40,000 h	[34,45]	1.37
5	Electrolyser (PEM-EL)	1500/kW	1000/kW	20/kW	660/kW	15	[38,45,48]	1.37
6	Low Pressure Tank	665/kg	400/kg	3/kg	260/kg	15	[38,44,45]	1.08
7	High-Pressure Tank	600/kg	400/kg	3/kg	80/kg	25	[38,44,45]	1.08
8	Compressor	2500/kW	2500/kW	50/kW	0	10	[13,45]	1.29
9	Water Pump	500/kW	500/kW	50/kW	0	10	[37,45]	1.29
10	Pre-cooler	5000/unit	5000/unit	0	3330/unit	15	[38,44,45]	1.08
11	Dispenser	54000/unit	54000/unit	0	0	10	[44,45]	1.19
12	Water Tank	200/unit	200/unit	0	0	10	[45,49]	1.08

Table 3

Solar PV, Wind Turbine and General Project Parameters used in the case study [46,50,51].

Solar PV Parameters		Wind Turbine Parameters		General Project Parameters	
Area of Solar Panel (m^2)	1.75	Cut-in wind speed (m/s)	2.7	Project Lifetime (years)	20
Nominal Efficiency	18.03 %	Cut-out wind speed (m/s)	9	Discount rate (%)	8.25
Temperature co-efficient (1/K)	−0.45 %	Rated windspeed (m/s)	27	Factor of safety	1.2
NOCT (K)	318 ± 2	Rated Pwer (kW)	50	Discount rate uncertainty set (%)	3.5—23.99
Test condition temperature (K)	298	Wind speed overall uncertainty set (m/s)	0.03—19.33		
Nominal Power of solar PV (W)	350				
Solar irradiation overall uncertainty set (W/m^2)	0—1186.48				
Ambient temperature overall uncertainty set (K)	268.16—312.25				

Reference condition: Irradiance $1000 W/m^2$, Module Temperature $25 ^\circ C$, Air Mass 1.5.Nominal Operation Cell Temperature (NOCT): Sun $800 W/m^2$; Air $20 ^\circ C$; Wind speed 1 m/s.**Fig. 2.** Hourly solar irradiation data.

approximately 9 L and the amount of water needed for an hour is about 180 L which will require 1 kW rating water pump [37]. Therefore, 2000 L capacity of water tank is used in this study.

2.12. Model assumptions

The following assumptions are made in the modelling of the integrated energy system.

1. The pre-cooling unit unit is attached to a dispenser, and a single hose dispenser is needed for every 250 kg daily demand [36].
2. The flow of hydrogen in the lines between the electrolyser and the dispenser is adiabatic and steady.
3. A single type of solar PV (monocrystalline) is used in the study.
4. The carbon emissions (in tonnes) avoided for the all three cases are calculated for the only operational time.
5. The relationship between the salvage value and number of years of operation of components is linear [38].

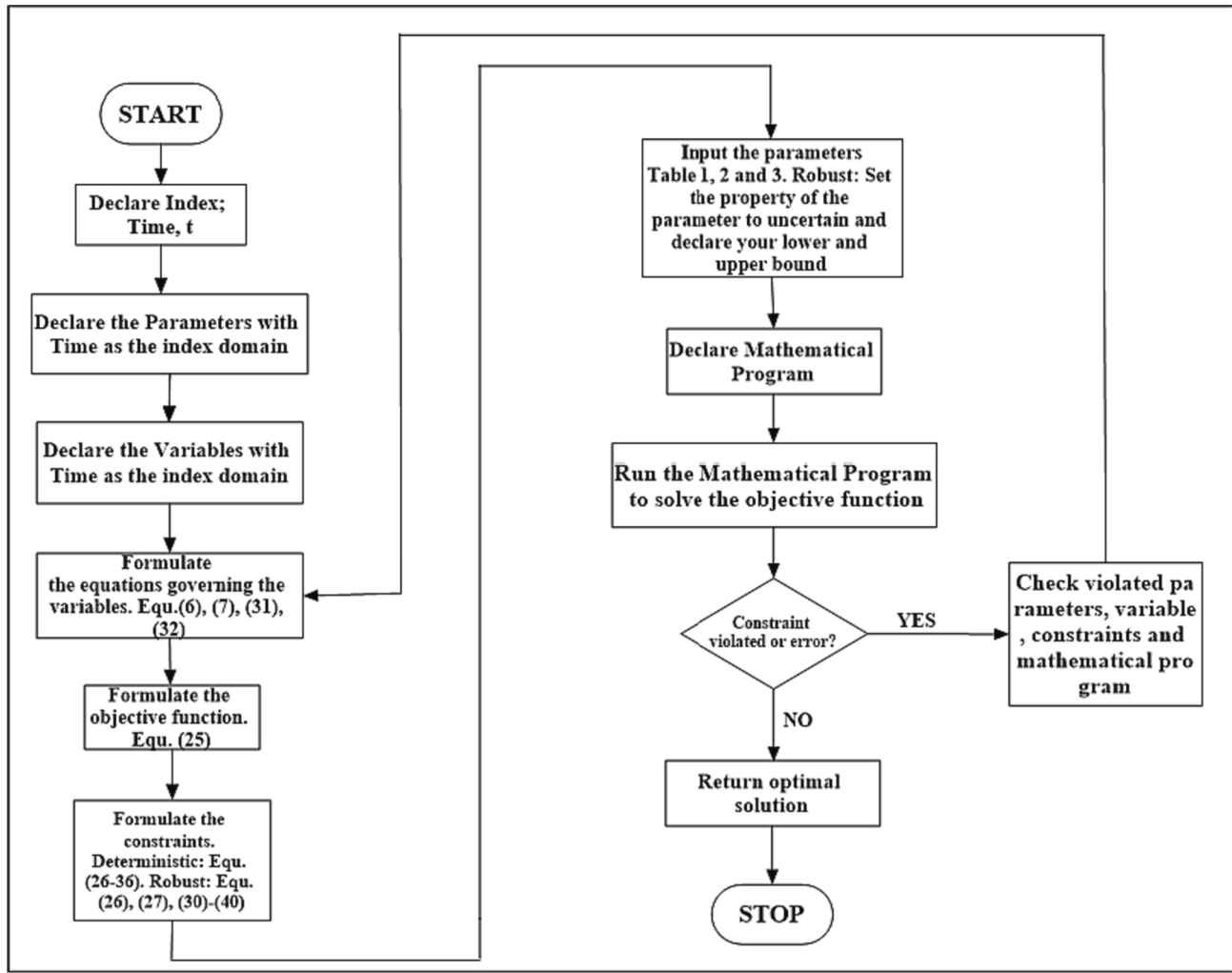


Fig. 3. Flow chart of the optimisation model.

2.13. Economic analysis

The total cost of the system and levelised cost of energy are computed using Equ. (17)–(24).

$$\text{Total Lifetime Cost (\$)} = IC + OM_{NPV} + R_{NPV} - S_{NPV} \quad (17)$$

$$LCOE (\$/kWh) = \frac{\text{Total Lifetime Cost (\$)}}{\text{Total Lifetime Energy Generated (kWh)}} \quad (18)$$

$$IC = P_R * IC_0 * CEPCI \quad (19)$$

$$OM_{NPV} = \sum_{t=1}^N \frac{P_R * OM(/year)}{(1+r)^t} \quad (20)$$

$$R_{NPV} = \frac{P_R * R(/unit)}{(1+r)^n} \quad (21)$$

$$S_{NPV} = \frac{S}{(1+r)^n} \quad (22)$$

$$\begin{aligned} \text{Salvage Value} &= \text{Purchase Price} \\ &- (\text{Annual Depreciation Cost} * \text{Number of Useful Life}) \end{aligned} \quad (23)$$

$$\text{Annual Depreciation Cost} = \frac{\text{Total Capital}}{\text{Lifecycle}} \quad (24)$$

A detailed financial analysis is carried out to obtain the benefits associated with the investment. All costs and potential revenue streams are converted to the present time to evaluate the profitability of the energy system. Through this analysis, the payback period and net present value of the different configurations considered in the study are evaluated. Based on the study in [39], the selling price of hydrogen is fixed at 6 \$/kg and based on the average electricity tariff in South Africa [40], the selling price of electricity is fixed at 0.1414 \$/kWh. The analysis incorporates the assumption of an annual energy price increase of 4 % and a drop in generated energy of 2 %. Where IC , OM_{NPV} , R_{NPV} , S_{NPV} , P_R , IC_0 , $OM(/year)$, $R(/unit)$, r , N , n and NPV are the initial cost, the maintenance cost, replacement cost, salvage value, rated power/capacity of components, capital cost per unit, maintenance cost per unit, replacement cost per unit, interest rate, project life time, referenced year and net present value respectively. CEPCI denotes chemical engineering plant cost index. The CEPCI includes the installation cost, planning, and other costs required for each part of the equipment. The total lifetime cost will be expressed in terms of five variables which are number of PVs, number of wind turbines, electrolyser capacity (kW), fuel cell capacity (kW) and capacity of the low-pressure tank (kg). The capacities of cascade, compressor, pre-cooler, converter, dispenser, water pump and water tank are known because they depend on the hourly demand of hydrogen which is a parameter. The addition of total cost of known capacities forms the constant term in the total cost function.

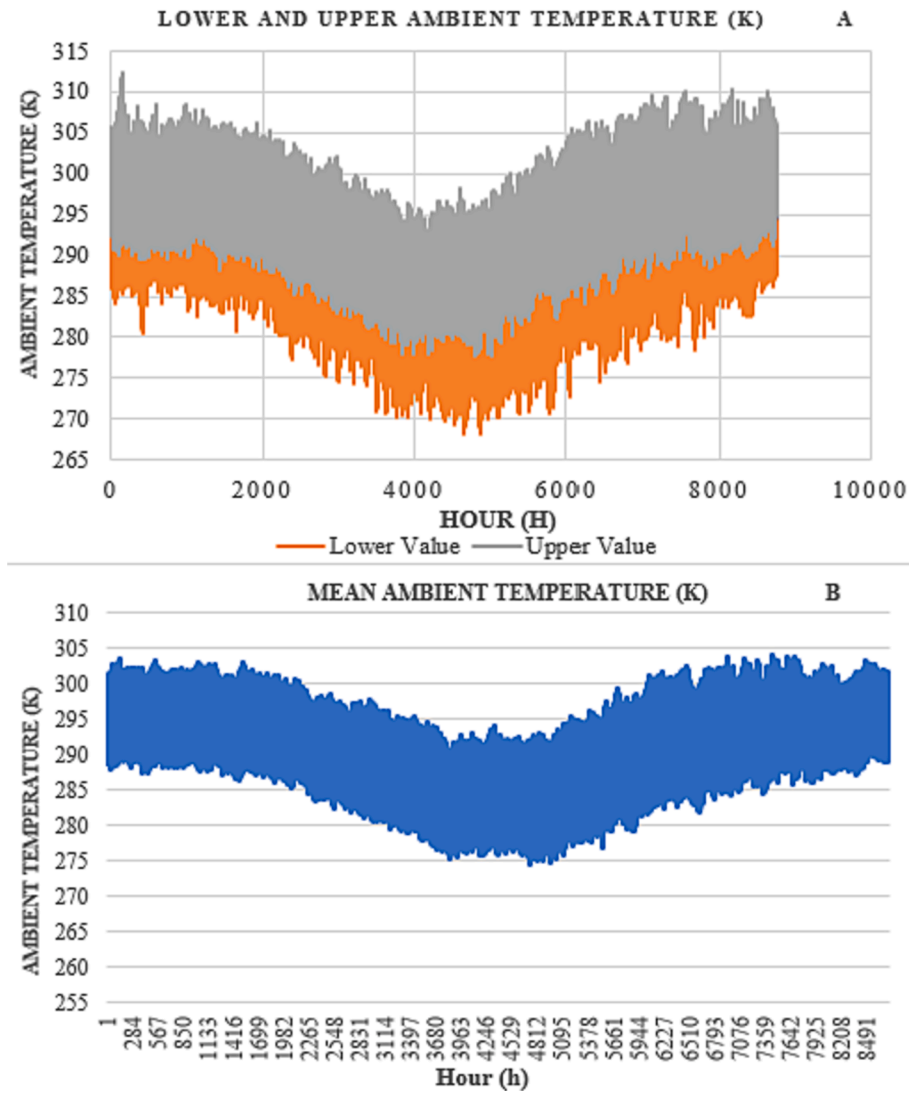


Fig. 4. A. Lower and Upper hourly ambient temperature data, B: Mean hourly ambient temperature data.

2.14. Carbon footprint & carbon tax

One of the aims of this study is to investigate the potential impact of utilising renewable energy sources on the reduction of greenhouse gas emissions within the transportation and power sectors. Within the transportation sector, green hydrogen has the potential to serve as a viable substitute for petrol and diesel, while the utilisation of renewables in conjunction with storage technologies holds promise for replacing coal-fired power plants. Several modelling exercises were undertaken in order to ascertain the optimal carbon tax rate in South Africa and the penalty is set at the base value of \$8 per tCO₂e [41]. Table 1 gives the equivalence of CO₂ for gasoline and coal.

2.15. Optimisation model formulation

The deterministic optimisation was carried out using AIMMS Solver ODH-CPLEX 5.3. AIMMS is a prescriptive analytics software for optimising mathematical models and very useful in supply chain management. The objective function of the optimisation model is total cost of the different configurations of the system (Solar PV & Wind Turbine, Solar PV, Wind Turbine). The decision variables are N_{pv} , N_{wt} , P_{el} , P_{fc} and $m_{\max}$.

Objective Function

$$\text{Minimise}[TotalLifetimeCost(N_{pv}, N_{wt}, P_{el}, P_{fc}, m_{\max})] \quad (25)$$

Constraints

$$N_{pv} * P_{pv}(t) + N_{wt} * P_{wt}(t) + P_{fc}(t) = \frac{L(t)}{\eta_{conv}} + P_{el}(t) \quad (26)$$

$$L(t) = E_{load}(t) + P_{comp}(t) + P_{pcu}(t) \quad (27)$$

$$P_{el}(t) \leq X_{el}(t) * P_{el} \quad (28)$$

$$P_{fc}(t) \leq X_{fc}(t) * P_{fc} \quad (29)$$

$$X_{el}(t) + X_{fc}(t) \leq 1 \quad (30)$$

$$m_{H_2d}(t) = \frac{P_{fc}(t)}{\eta_{fc} * HHV_{H_2}} \quad (31)$$

$$m_{H_2a}(t) = \frac{P_{el}(t) * \eta_{ele}}{HHV_{H_2}} \quad (32)$$

$$m(t) = m(t-1) - H_d(t) + m_{H_2a}(t) - m_{H_2d}(t) \quad (33)$$

$$m_{\min} = 0.05 * m_{\max} \quad (34)$$

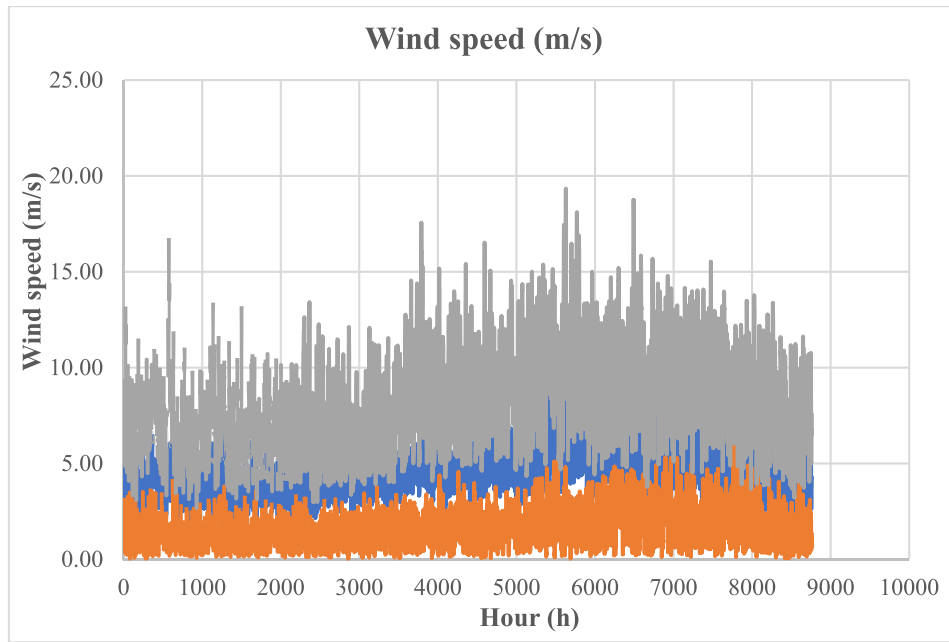


Fig. 5. Hourly wind speed data at 70 m hub height.

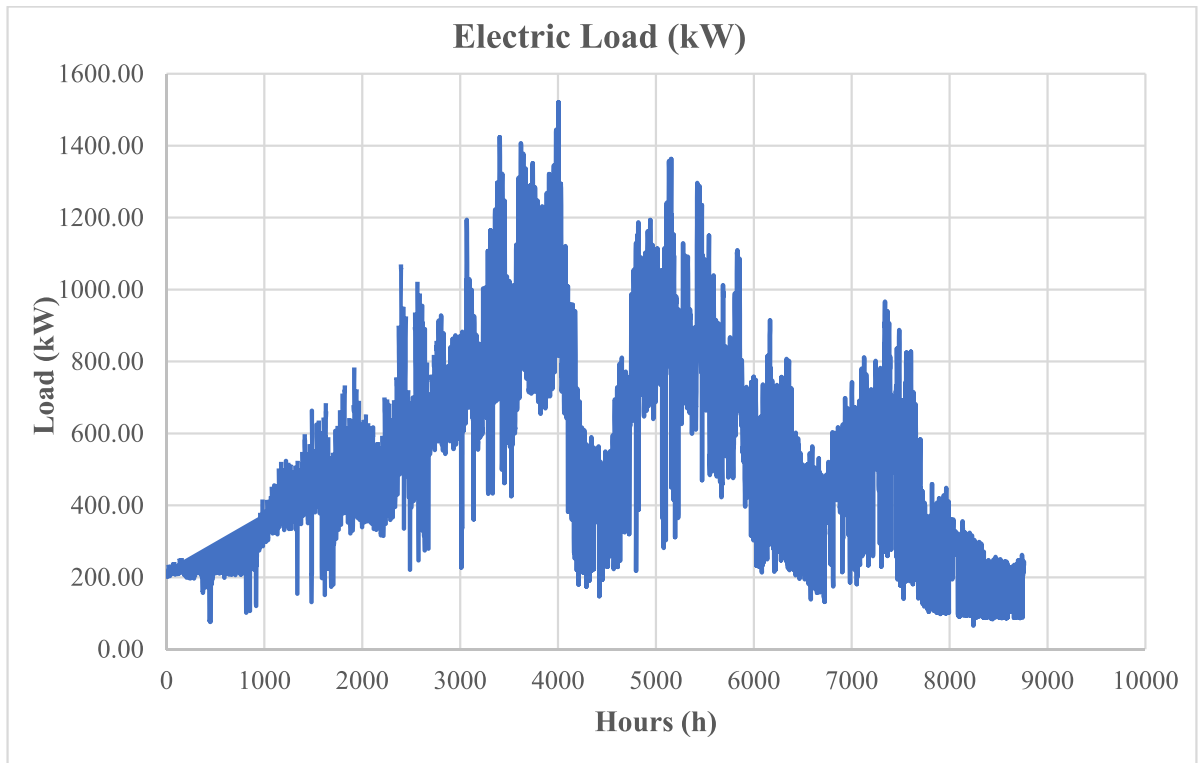


Fig. 6. Hourly electric load data.

$$m_{\min} \leq m(t) \leq m_{\max} \quad (35)$$

$$X_{el}(t), X_{fc}(t) \in (0, 1) \quad (36)$$

The constraints Equ. (28) and Equ. (29) makes the optimisation model MIQCP. The two equations are further relaxed using Big M to Equ. (37)–(40) resulting in MILP. The MILP is necessary to run the robust optimisation in AIMMS. AIMMS has robust optimisation add-on which enables the user to carry out optimisation technique under uncertainty

conditions. The essential steps for constructing a robust optimisation model involve explicitly identifying the parameters in the deterministic model that are intended to be treated as uncertain, specifying the variables in the deterministic model that are intended to be adjustable to the uncertain parameters (if applicable), and providing a clear specification of the potential realisations of the uncertain parameters [43].

$$P_{el}(t) \leq X_{el}(t) * P_{\max} \quad (37)$$

$$P_{fc}(t) \leq X_{fc}(t) * P_{\max} \quad (38)$$

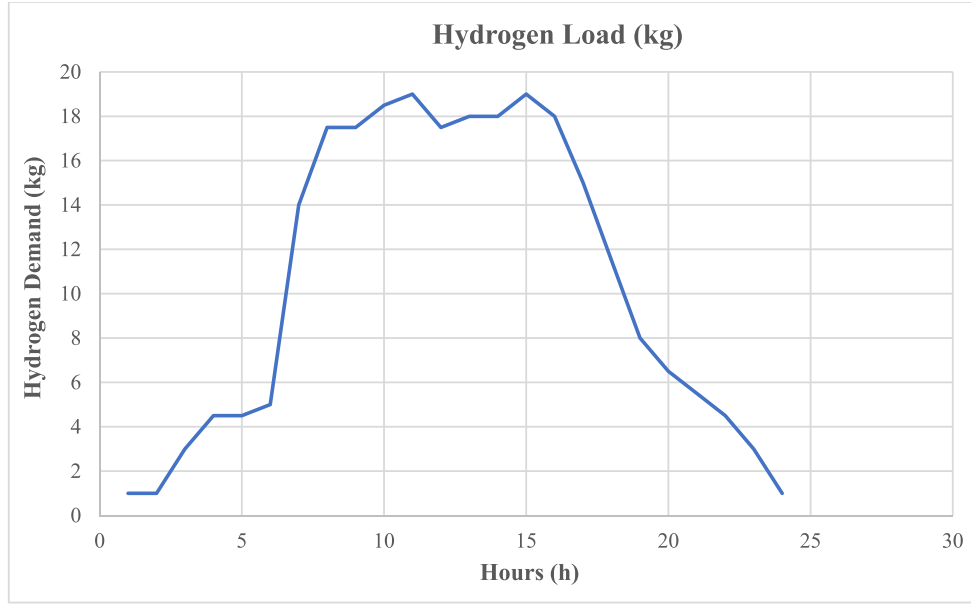


Fig. 7. Hourly hydrogen load data.

Table 4

Optimised technical variables of the integrated energy system for the considered configurations for deterministic solution.

S/ N	Components	Hybrid Solar PV and Wind Turbine System	Standalone Solar PV	Standalone Wind Turbine
1	Number of wind turbine	610	0	690
2	Number solar PV capacity	5,660	358,100	0
3	Capacity of the wind farm (MW)	30.5	0	34.5
4	Capacity of the solar farm (MW)	2	125	0
5	Converter (kW)	3,170	6,760	3,535
6	Fuel Cell Capacity (kW)	1,280	1,625	1,295
7	Electrolyser Capacity (kW)	2,270	4,830	2,525
8	Low Pressure Tank Capacity (kg)	770	770	895
9	Cascade (kg)	25	25	25
10	Compressor (kW)	70	70	70
11	Water Pump (kW)	1	1	1
12	Pre-cooler (kW)	6.5	6.5	6.5
13	Dispenser (unit)	1	1	1
14	Water Tank (unit)	1	1	1
15	LCOE (\$/kWh)	0.0783	0.0702	0.0786
16	Carbon Emissions Avoided (Tonnes)	928,500	5,878,730	952,160
17	Carbon Tax (\$)	7,427,900	47,029,800	7,617,300
18	Total Excess Energy (GWh)	28	256	29

Table 5

Total cost of the integrated energy system for the considered configurations for deterministic solution.

S/ N	Components	Hybrid Solar PV and Wind Turbine System	Standalone Solar PV	Standalone Wind Turbine
1	Wind Turbine	45,238,280	0	51,552,330
2	Solar PV	5,620,860	355,496,880	0
3	Converter	3,652,650	7,786,810	4,072,880
4	Fuel Cell	6,350,300	8,084,700	6,441,830
5	Electrolyser	5,729,140	12,213,550	6,388,270
6	Low Pressure Tank	637,950	636,740	739,170
7	Cascade	15,500	15,500	15,500
8	Compressor	356,850	356,850	356,850
9	Water Pump	1,420	1,420	1,420
10	Pre-cooler	49,180	49,180	49,180
11	Dispenser	93,350	93,350	93,350
12	Water Tank	320	320	320
	Total (\$)	67,745,800	384,735,300	69,711,100

cell (kW), $m_{\max}$ is the maximum capacity of the low pressure tank (kg), $m_{\min}$ is the minimum capacity of the low pressure tank (kg), $P_{pv}(t)$ is the hourly power output from one solar PV, $P_{wt}(t)$ is the hourly power output from a wind turbine, $P_{fc}(t)$ is the hourly power supplied from a fuel cell (kW), $P_{el}(t)$ is the hourly power supplied to the electrolyser (kW), $L(t)$ is the total hourly load (kW), $E_{load}(t)$ is the hourly electrical load (kW), $P_{comp}(t)$ is the hourly compressor load (kW), $P_{pcu}(t)$ is the hourly pre-cooler load (kW), $X_{el}(t)$ is the binary determinant for electrolyser operation, $X_{fc}(t)$ is the hourly binary determinant for fuel cell operation, $H_d(t)$ is the hourly hydrogen load (kg). Equ. (26) states the objective function of the optimisation model. The total cost of the system is to be minimised. The decision variables are number of PVs, number of wind turbines, electrolyser capacity (kW), fuel cell capacity (kW) and capacity of the low-pressure tank (kg). The objective of the optimisation is to minimise the system's total cost. Equ. (27) explains power balance at every time step. The excess energy is used to power the electrolyser to store hydrogen in the low-pressure tank whenever there is surplus. The fuel cell is powered to supply the shortfall through the hydrogen in the low-pressure tank whenever there is deficit. The cascade takes hydrogen from the low-pressure tank to meet the hydrogen demand at any time. Invariably, Equ. (27) states that the load must be met at all times which means that the reliability of the system to meet the load is maximum.

$$0 \leq P_{el}(t) \leq P_{el} \quad (39)$$

$$0 \leq P_{fc}(t) \leq P_{fc} \quad (40)$$

The $P_{\max}$ is a parameter decided to set a ceiling for $P_{el}(t)$ and $P_{fc}(t)$. Where N_{pv} is the number of solar panels, N_{wt} is the number of wind turbines, P_{el} is the capacity of electrolyser (kW), P_{fc} is the capacity of fuel

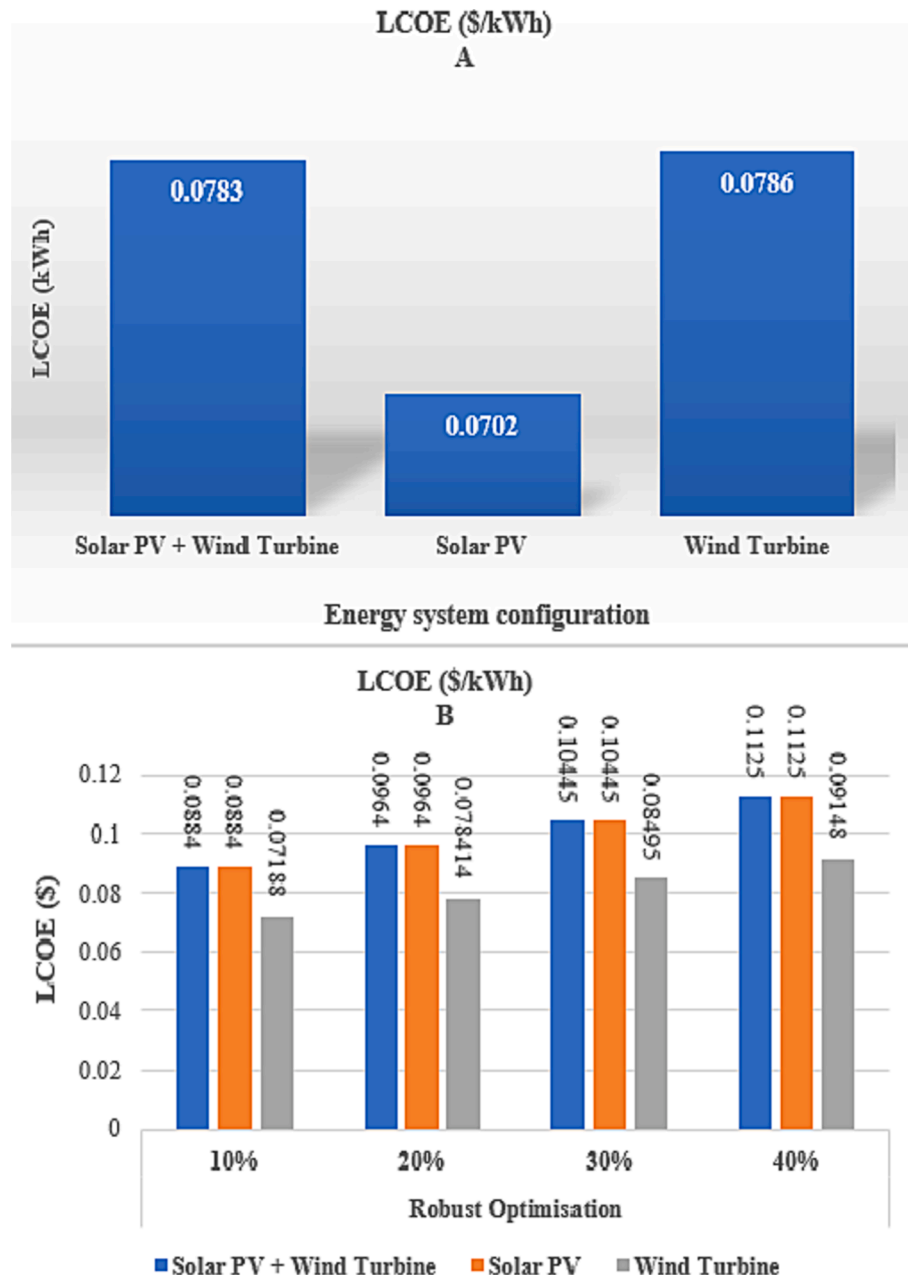


Fig. 8. Levelised cost of energy of the configurations for the deterministic and robust solution.

The compressor and dispenser/pre-cooler are also needed to be powered to operate. Equ. (28) explains that the total load is the summation of electrical load, compressor, and dispenser/pre-cooler load. Equ. (29)–(31) states that the operation of electrolyser and fuel cell are mutually exclusive. The electrolyser and fuel cell cannot operate simultaneously. Either one of the two components is operating at a time, or none is working and in the event that none of them is operating, the power sources provide just the energy needed for the load. The fuel cell power only serves the load, it will not be used to power the electrolyser. Equ. (32)–(34) explains the mass balance relation of the low-pressure tank at any time step. Equ. (35)–(36) explains the low-pressure tank limit. It is recommended that the minimum mass of hydrogen in the tank be set at 5 % because there is always a decline in pressure when a small portion of hydrogen is left in the tank [17,24]. Table 2 gives the cost parameters used in the study while Table 3 gives the technical parameters of the wind turbine and solar PV with the general project parameters.”.

3. Data and uncertainty quantifications

Solar Irradiation and wind speed are highly unpredictable. Hence, in order to obtain the uncertainty data sets for the irradiation, ambient temperature and wind speed for the location under consideration (Lat. -26.1847 , Long. 27.9974), the historical solar irradiation, ambient temperature and wind speed data of the city of Johannesburg, South Africa from the year 2011 to 2021 was obtained from Merra – 2 Meteorological Re-Analysis [52] and analysed. The minimum and maximum values for each time steps over the range of years were taken as the floor and ceiling of the uncertainty data set of solar irradiation, ambient temperature, and wind speed. Furthermore, unlike most studies that perform economic analysis under a fixed discount rate, which often lead to poor performance, this study used the limits the discount rate in South Africa has reached as published by Reserve Bank [53]. The discount rate’s minimum and maximum levels form the uncertainty set’s floor and ceiling. Furthermore, the uncertainties in the costs of each

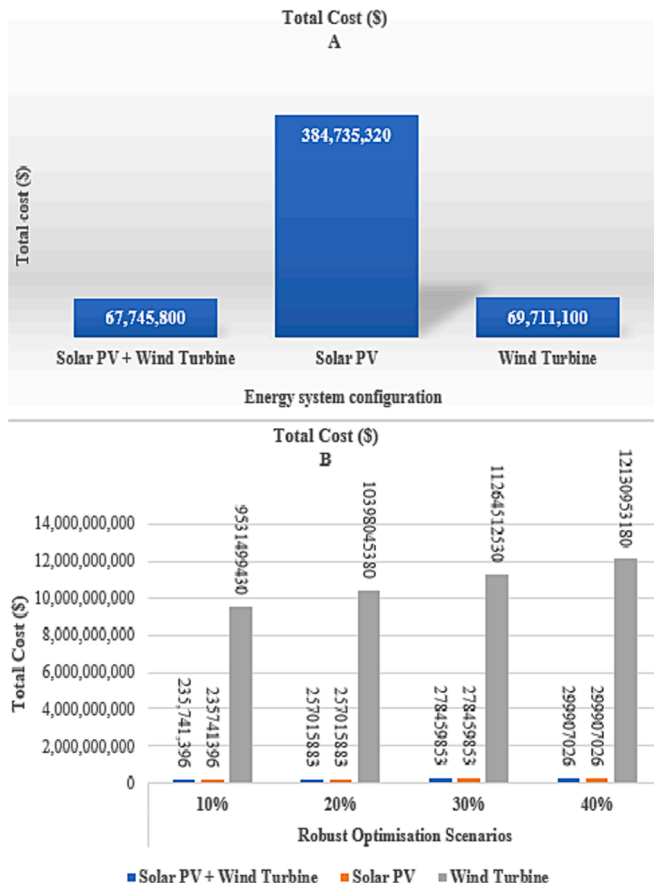


Fig. 9. Integrated Energy systems' total cost for the deterministic and robust solution.

component are set at 10 %, 20 %, 30 % and 40 % upper and lower limit to ascertain the variations in the variables being investigated using sensitivity analysis. The electricity consumption of the halls of residence of the University of Johannesburg, Auckland Park, South Africa is used as the electric load and refuelling station load is taken to be 250 kg daily. The assumed hourly demand pattern for hydrogen refuelling stations in South Africa is expected to resemble the existing pattern observed in conventional refuelling stations. This assumption is supported by the fact that one of the earliest sets of hydrogen refuelling stations at the University of California in Irvine has been operating since 2002 and follows a similar demand pattern[54,55]. Fig. 2 illustrates the hourly solar irradiation. Fig. 3 displays the chart of the optimisation process. Fig. 4A depicts the lower and upper hourly ambient temperature while Fig. 4B depicts the hourly mean ambient temperature, Fig. 5 illustrates the hourly speed data, Fig. 6 illustrates the electric load data and Fig. 7 illustrates the hourly hydrogen load data. In summary, the uncertainties exhibiting parameters are the windspeed, solar irradiation, ambient temperature, unit costs and discount rate.

4. Numerical simulations

The study establishes the practicability of hydrogen vector produced from an integrated energy system for mobility and storage applications. In order to meet both the electric load and the hydrogen load simultaneously, the optimal sizes of the components of the integrated system of hydrogen refuelling station and photovoltaic - wind turbine hydrogen-based storage are determined. The system was modelled using MIQCP and further relaxed to MILP to evaluate its economic, technical, and environmental benefits. ODH CPLEX 5.3 solver was used for the deterministic optimisation while CPLEX 22.1 solver was used for the robust optimisation. The overall uncertainty set for the parameters under perturbation are presented in Table 3. Fig. 6 and Fig. 7 refers to the hourly forecasted electricity and hydrogen load of the system. The annual electric load and hydrogen load to be met are 4.7 GWh and 91.3 tonnes respectively. The hourly environmental data and their uncertainties are depicted in Figs. 2, 4 and 5. Based on the model in Equ. (1)–(3),

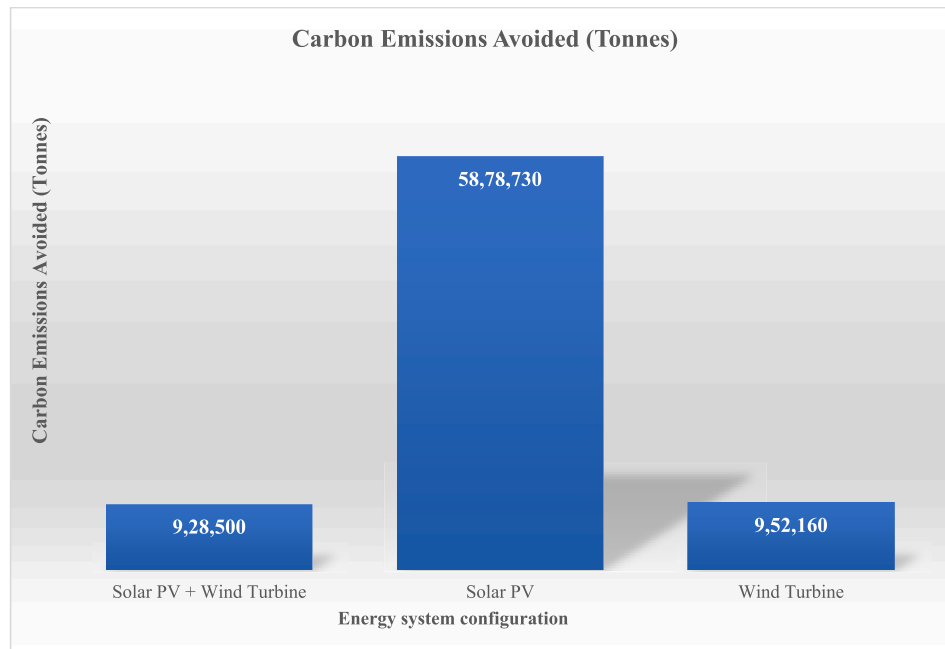


Fig. 10. Carbon emissions saved by the integrated energy system.

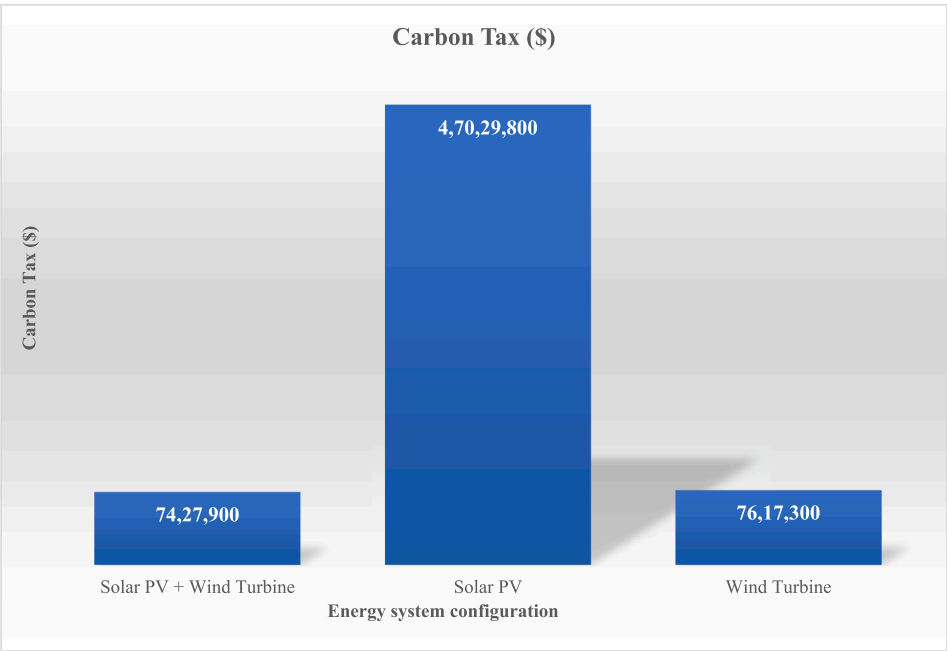


Fig. 11. Carbon taxes saved by the integrated energy system.

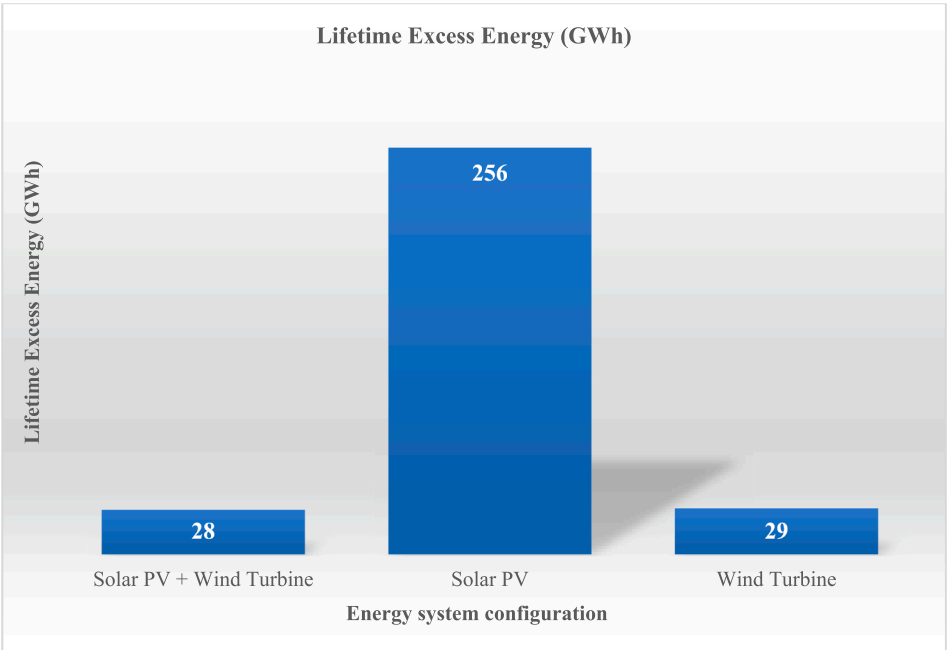


Fig. 12. Excess energy from the different configurations of the system.

temperature and solar irradiation profile, the power harvested by the solar PV is computed. The same principle applies for the wind turbine, Equ. (5) extrapolates the measured wind speed at 10 m to the hub height of 70 m, while Equ. (4) computes the generated power based on the wind turbine power curve.

4.1. Results

This section evaluates the results obtained from the deterministic and robust optimisation of the integrated energy system for three configurations namely, combined wind and solar PV, solar PV and wind turbine. Table 4 and 5 shows the sizes of the sub-systems, the total cost,

cost of each component, levelised cost of energy, excess energy available to be exported to the grid, equivalent mass of CO₂ saved and carbon tax avoided by the integrated energy system based on deterministic optimisation. Figs. 8, 9, 10, 11, 12, 13 and 14 display the deterministic and robust solution LCOE, deterministic and robust solution total cost, carbon emissions saved, carbon tax saved by the configurations, excess energy, the net present value of the configurations and the net cash flow diagram respectively. Considering the robust optimisation solution, sizes of each component for the different system configurations, the LCOE, total excess energy and total cost are presented in Table 6 and Table 7. The goal of the model for both deterministic and robust optimisation is to investigate all the configurations based on the total cost.

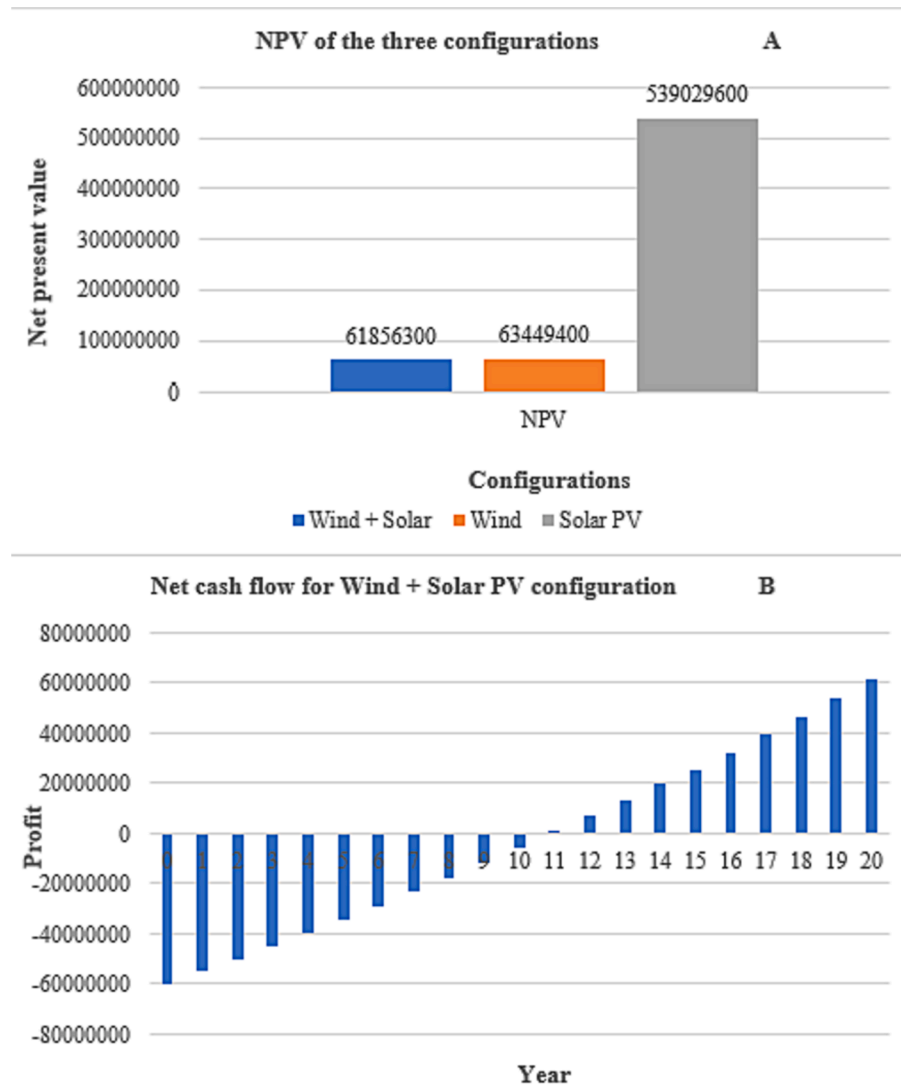


Fig. 13. A: Net present value of the three configuration, B: Net cash flow diagram of solar and wind configuration.

4.2. Discussions

Discussion on the results provided in section 4.1 are as follows.

4.3. Component sizing of the three configurations of the integrated energy system

The total cost optimisation models were used to determine the sizes of the components of the three configurations. From the deterministic and robust optimisation model solution, number of solar PV, number of wind turbine, power rating of converter, capacity of fuel cell, capacity of electrolyser and capacity of low-pressure tank are presented in table Table 4 and Table 6. The sizes of other components, cascade (kg), compressor (kW), water pump (kW), pre-cooler (kW), dispenser (unit) and water tank (unit) are evaluated based on the hourly hydrogen demand of the refuelling station and other parameters which have been highlighted in sections 2.7 to 2.11. The optimisation model constrains the system to meet both the electric and hydrogen demands at all times as represented in Equ. (26). Therefore, the sizes obtained ensures 100 %

reliability performance of the system. All the components can be generally classified in to three categories; The solar PV and wind turbine are the energy generators, the electrolyser, fuel cell, low-pressure tank and cascade serve as the storage medium while the converter, compressor, water pump, pre-cooler, dispenser, and water tank are the supporting components for power production and hydrogen refuelling. In terms of the compactness (required space and power rating) of the components of the integrated energy, the hybrid and standalone wind turbine configuration is a more suitable option.

4.4. Total cost and levelised cost of energy

For both optimisation cases, the total cost for each configuration was minimised and the corresponding levelised cost of energy was obtained.

From the results of the deterministic solution, the hybrid and standalone wind energy system configuration is a more suitable option in terms of the total cost of the system while the standalone solar PV system has the least levelised cost of energy. The levelised cost of energy is dependent on both the total cost and total energy produced.

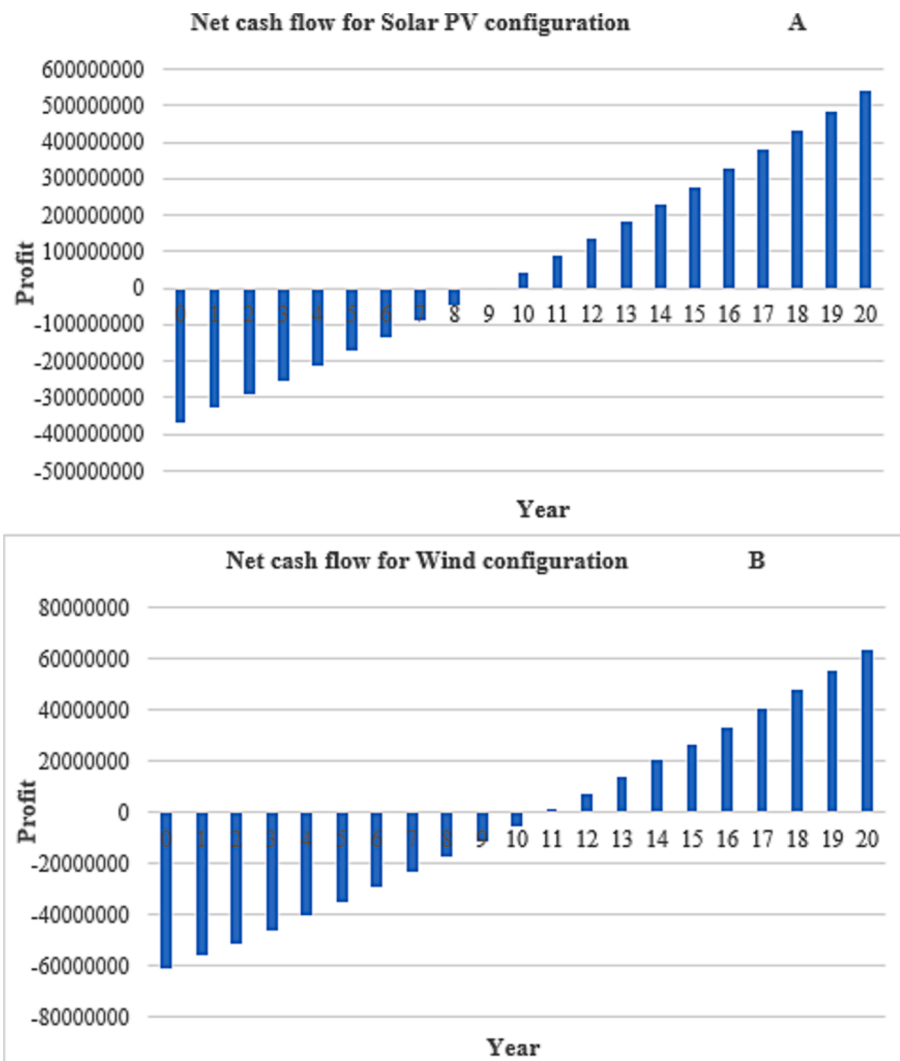


Fig. 14. A: Net cash flow diagram of solar configuration, B: Net cash flow diagram of wind configuration.

Standalone solar PV has the least levelised cost of energy because of its higher differential amount of energy generated throughout its lifetime. The solar irradiation is not always available and varies throughout the day, hence there is need for enough photovoltaic panels and storage media to capture, convert, and store energy when available. This often leads to large number of solar PVs, large capacities of converter/electrolyser/fuel cell and abundant excess energy. In this case, the scale of the system is higher than other configurations leading to lower levelised cost of energy. Hypothetically, the excess energy is not curtailed but fed to the South African grid increasing the revenue of the system of the system and reducing the payback period and internal rate of return of the project. For the robust solution, the hybrid and standalone solar PV is more suitable in terms of total cost of the system while the standalone wind energy system has the least levelised cost of energy. The low wind energy at the lower bound of the wind speed contributes to the high rated power of the wind turbine which affects the total cost of the standalone wind energy system and excess energy. This calls for a greater number of wind turbines to generate the required amount of power. It is also observed that the levelised cost of energy increases as perturbation increases showing the conservative property of robust optimisation. It can also be observed from the breakdown of the cost of components that the renewables take a huge fraction of the overall cost, consequently, to reduce the total cost and levelised cost of energy, the cost of renewables must further be reduced. The highest levelised cost of energy obtained in this study is 0.1125 \$/kWh. C.A.W Ngouleu et al. [9]

performed a cost analysis and optimisation sizing of hybrid renewable system with battery and hydrogen storage for rural locations in Cameroun. The range of levelised cost of energy obtained is 0.1959 \$/kWh – 0.2386 \$/kWh. A. Khiareddine et al. [20] conducted a study on sizing methodology for combined photovoltaic-wind-hydrogen-battery integrated for pumping system for a location in Tunisia. The range of levelised cost of energy obtained is 1.064 \$/kWh – 8.74 \$/kWh. Coppitters et al. [15] proposed a grid-connected solar PV with battery/hydrogen storage for a location in Belgium and was optimised using multiple scenarios of robust optimisation and then compared with stochastic performance based on the corresponding cumulative density functions. The range of levelised cost of energy obtained is 0.269 \$/kWh – 0.404 \$/kWh. M.M. Samy et al. [17] investigated a techno-economic study of solar PV-wind-hydrogen storage with unreliable grid using a hybrid search optimisation technique for a tourist resort in Egypt. The range of levelised cost of energy obtained is 0.0456 \$/kWh – 0.1443 \$/kWh. C. Ghenai et al. [8] presented a modelling, simulation and optimisation of solar PV and fuel cell to meet the desired electrical load of 150 houses in the Emirate of Sharjah, UAE. The levelised cost of energy obtained is 0.145 \$/kWh. The average electricity tariff in South Africa is 0.1414 \$/kWh [40]. In general, these unit costs are deemed competitive in relation to the energy generation costs from other regions in the world.

Table 6

Optimised technical variables of the integrated energy system for the considered configurations for robust solution.

S/N	Components	Hybrid Solar PV and Wind Turbine System	Standalone Solar PV	Standalone Wind Turbine
1	Number of wind turbine	0	0	103,740
2	Number solar PV capacity	174,250	174,250	0
3	Capacity of the wind farm (MW)	0	0	5,187
4	Capacity of the solar farm (MW)	61	61	0
5	Converter (kW)	8,250	8,250	192,140
6	Fuel Cell Capacity (kW)	1,625	1,625	1,625
7	Electrolyser Capacity (kW)	5,895	5,895	137,250
8	Low Pressure Tank Capacity (kg)	760	760	22,250
9	Cascade (kg)	25	25	25
10	Compressor (kW)	70	70	70
11	Water Pump (kW)	1	1	1
12	Pre-cooler (kW)	6.5	6.5	6.5
13	Dispenser (unit)	1	1	1
14	Water Tank (unit)	1	1	1

Table 7

Levelised cost of energy and total cost of the integrated energy hub for the considered configurations for robust solution.

Level of Robust Optimisation	System Configuration	LCOE (\$/kWh)	Total Excess Energy (GWh)	Total Cost (\$)
Robust (10 %)	Solar PV + Wind Turbine	0.0884	60.92	235,741,400
	Solar PV	0.0884	62	235,741,400
	Wind Turbine	0.07188	258.2	9,531,499,450
Robust (20 %)	Solar PV + Wind Turbine	0.0964	60.91	257,015,880
	Solar PV	0.0964	61.91	257,015,880
	Wind Turbine	0.078414	287.4	10,398,045,380
Robust (30 %)	Solar PV + Wind Turbine	0.10445	99.18	278,459,850
	Solar PV	0.10445	61.91	278,459,850
	Wind Turbine	0.08495	283.8	11,264,512,530
Robust (40 %)	Solar PV + Wind Turbine	0.1125	99.19	299,907,000
	Solar PV	0.1125	61.91	299,907,000
	Wind Turbine	0.09148	273.61	12,130,953,180

4.5. Fossil-fuel replacement and carbon reduction capability of the integrated energy system

The carbon-based emissions from coal-fired power plant and gasoline cars could be avoided due to the replacement of these fossil-based energy systems with renewables. The solar-wind-hydrogen storage supplies the electric load while the green hydrogen refuelling station meets the hydrogen demand. Based on the information provided on CO₂ equivalent emission and carbon tax in [section 2.13](#), the amount of energy from the integrated energy system that could replace the coal-fired plant and gasoline cars was evaluated and the corresponding CO₂ equivalent emission and carbon tax was obtained. The results are detailedly presented in [Fig. 10](#), [Fig. 11](#) and [Table 4](#).

4.6. Excess energy

In instances where the electrical and hydrogen demands are met and the storage capacity is reached, any surplus energy generated by solar photovoltaic (PV) and wind turbine systems is allocated to various applications, including but not limited to heating, cooling, ventilation, desalination, water pumping, and power-to-gas conversion. This is particularly relevant in cases where the grid infrastructure is distant, rendering it economically unviable to transmit the excess energy to the grid. While in some scenarios, the excess energy is curtailed resulting in a prolonged payback period, the study proposes selling the excess energy to grid based on the time of use tariff and more revenue is obtained. This shortens the payback period and internal rate of return. The excess energy from the hybrid system, standalone solar PV system and standalone wind energy system are presented in [Table 4](#) and [Fig. 12](#). The injection of excess power to the increases the reliability of the grid and reduces load shedding and its impact in the country.

4.7. Net cash flow diagram

Detailed financial analysis carried out reinforces that all the three configurations are financially feasible based on the energy prices and conditions given in [section 2.13](#). The results are displayed in [Figs. 13](#) and [14](#). The payback period is within half of the project's lifetime while the net present value is positive. The payback period of the hybrid configuration and wind configuration is 11 years while that of the solar PV configuration is 10 years. Moreso, the result obtained for the net present value is related to the size or scale of the energy system. Solar PV configuration has almost 10x net present value than other configurations because of the scale.

5. Conclusion

This paper presents a technical and economic investigation of a combined system of electrical power generation and hydrogen refuelling station. Considering uncertainties in the wind resource, solar resource, costs and discount rate, the total cost is minimised to obtain the optimal configurations while the levelised cost of energy, excess energy, greenhouse emission avoided, and carbon tax were evaluated. Robust optimisation is used in this study to ensure that the results obtained are the worst-case possible scenarios. It considers the the highest costs, lowest environmental energy, and worst discount rate. The levelised cost of the deterministic optimisation solution ranges between 0.0702 \$/kWh to 0.0786 \$/kWh while the levelised cost of the robust optimisation solution ranges between 0.07188 \$/kWh to 0.1125 \$/kWh. 0.1125 \$/kWh is the highest levelised cost of energy obtained from this study for only solar PV and solar PV with wind turbine configuration which is cost-competitive to works conducted in previous literatures and energy tariffs for different regions in the globe. This study establishes the feasibility of producing electricity and hydrogen from a single infrastructure

with reduced cost and environmental benefits. This work will be further expanded in future with the inclusion of household water heating from thermal energy recovery from the solar PV, electrolyser, and fuel cell.

CRediT authorship contribution statement

Oladimeji Lawrence Oyewole: Conceptualization, Data curation, Data curation, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Resources. **Nnamdi Ikechi Nwulu:** Project administration, Supervision, Writing – review & editing. **Ewaoche John Okampo:** Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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